AWS VPC Peering

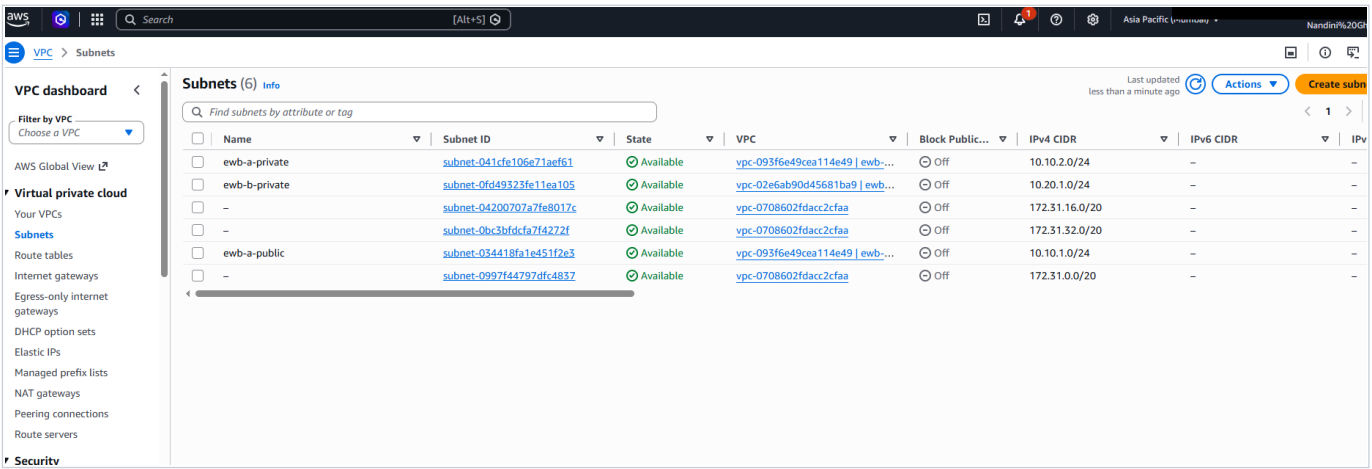
EWB Cloud Labs, Project 01

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Completion date	November 2026

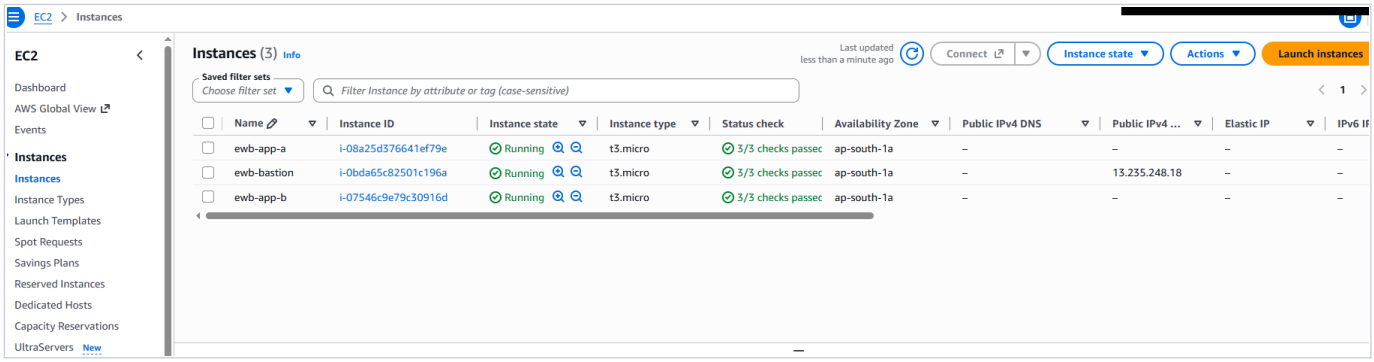
1. Architecture and inventory

Two VPCs with non-overlapping CIDRs are connected by a VPC peering connection. One public bastion in VPC A is the only host with a public IP. App-A and App-B stay private. Administration uses SSH ProxyJump through the bastion.

Resource	Name	CIDR / private IP	Actual ID
VPC A	ewb-vpc-a	10.10.0.0/16	vpc-093f6e49cea114e49
Public subnet	ewb-a-public	10.10.1.0/24	subnet-034418fa1e451f2e3
Private subnet A	ewb-a-private	10.10.2.0/24	subnet-041cfe106e71aef61
VPC B	ewb-vpc-b	10.20.0.0/16	vpc-02e6ab90d45681ba9
Private subnet B	ewb-b-private	10.20.1.0/24	subnet-0fd49323fe11ea105
Internet gateway	ewb-igw-a	Attached to VPC A	igw-064d7533e42e5f5c3
Peering connection	ewb-a-b-peer	VPC A to VPC B	pcx-0f5b68b9051af20a0
Bastion	ewb-bastion	10.10.1.10 + public IP	i-0bda65c82501c196a
App-A	ewb-app-a	10.10.2.10, no public IP	i-08a25d376641ef79e
App-B	ewb-app-b	10.20.1.10, no public IP	i-07546c9e79c30916d



Subnets list showing ewb-a-public, ewb-a-private and ewb-b-private with their CIDRs.

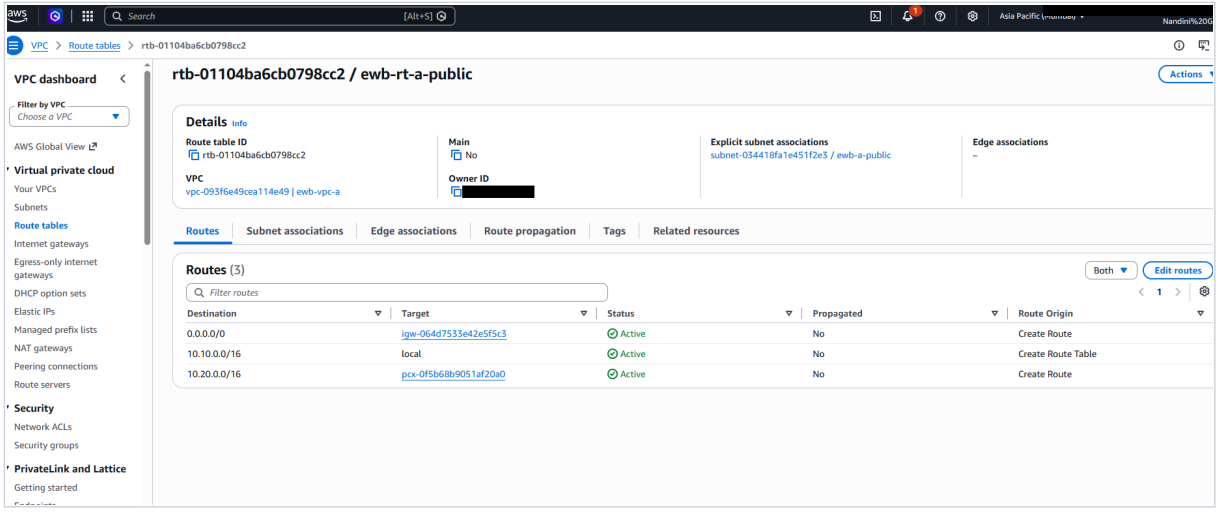


Instances list: ewb-bastion has a public IPv4 address; ewb-app-a and ewb-app-b have none.

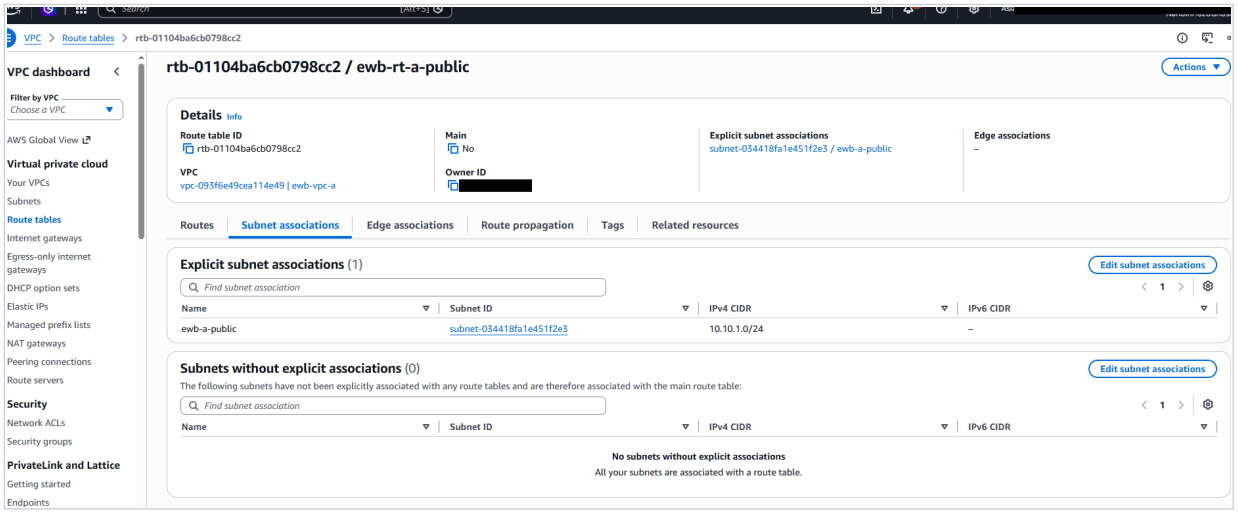
2. Route tables and subnet associations

Route table (associated subnet)	Destination	Target
ewb-rt-a-public (ewb-a-public)	10.10.0.0/16	local
	0.0.0.0/0	ewb-igw-a
	10.20.0.0/16	pcx (ewb-a-b-peer)
ewb-rt-a-private (ewb-a-private)	10.10.0.0/16	local
	10.20.0.0/16	pcx (ewb-a-b-peer)
ewb-rt-b-private (ewb-b-private)	10.20.0.0/16	local
	10.10.0.0/16	pcx (ewb-a-b-peer)

ewb-rt-a-public (rtb-01104ba6cb0798cc2)

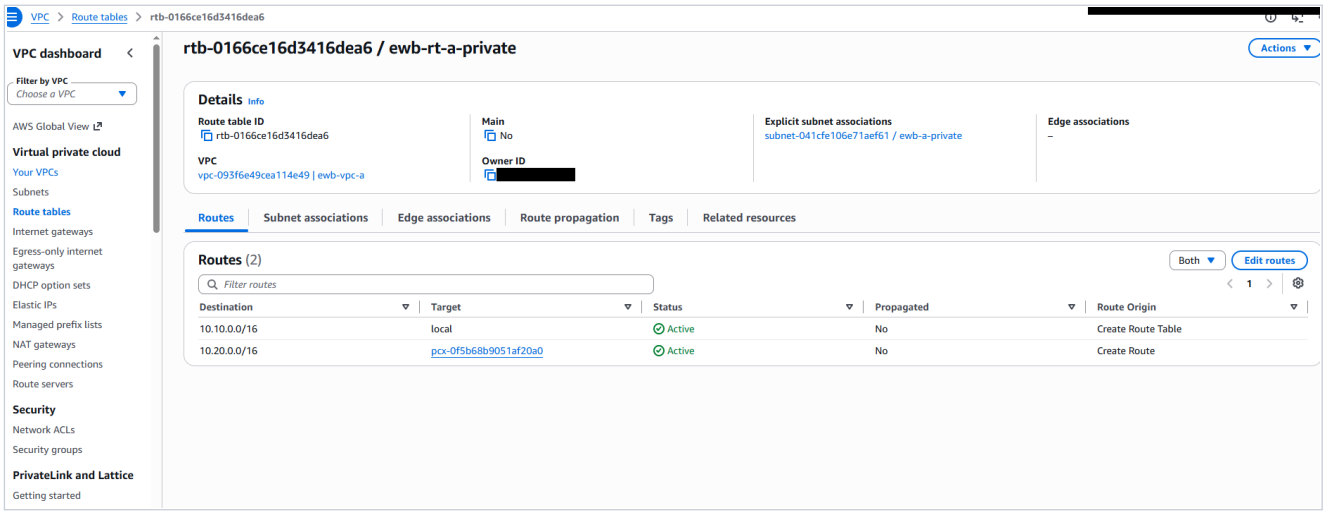


ewb-rt-a-public, Routes tab.

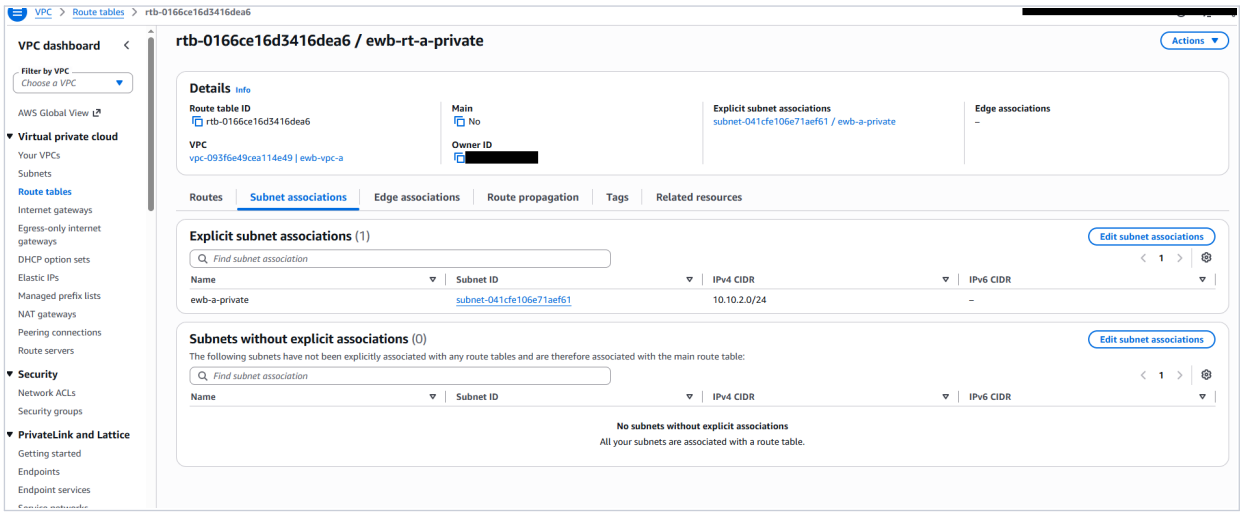


ewb-rt-a-public, Subnet associations tab.

ewb-rt-a-private (rtb-0166ce16d3416dea6)

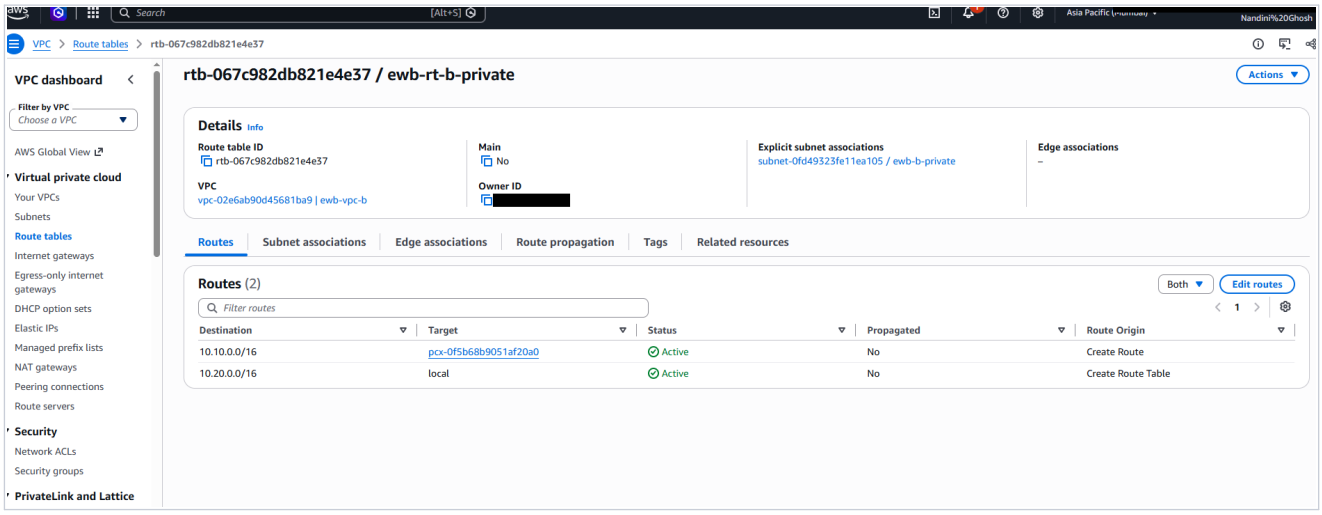


ewb-rt-a-private, Routes tab.

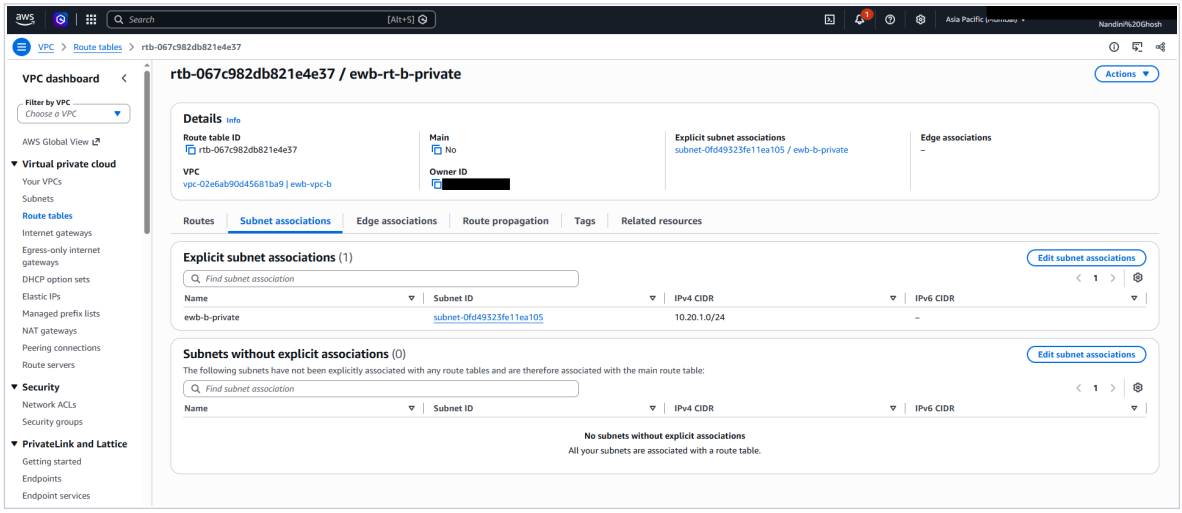


ewb-rt-a-private, Subnet associations tab.

ewb-rt-b-private (rtb-067c982db821e4e37)

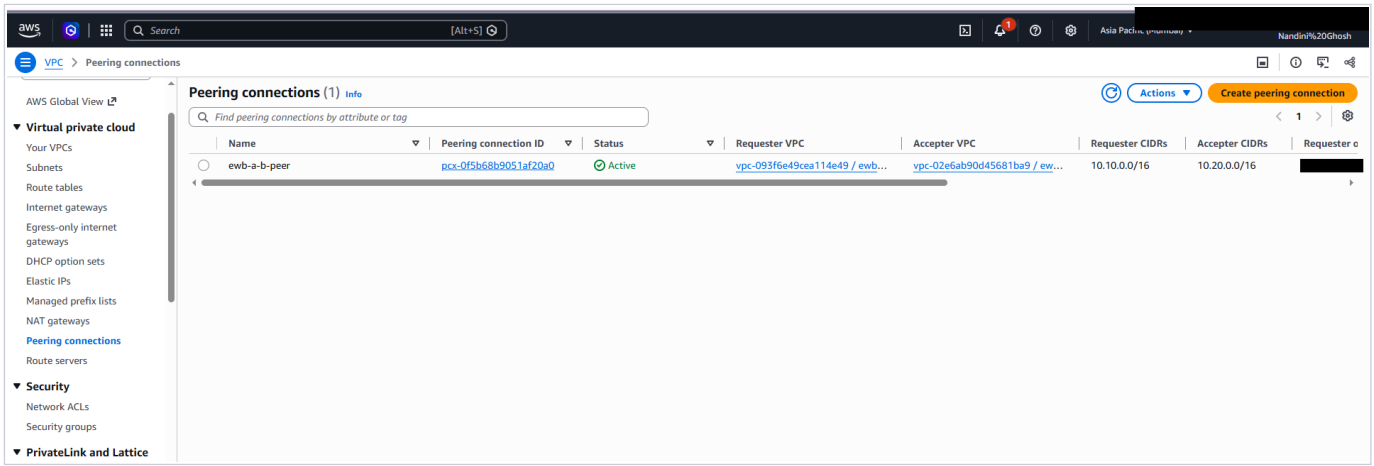


ewb-rt-b-private, Routes tab.

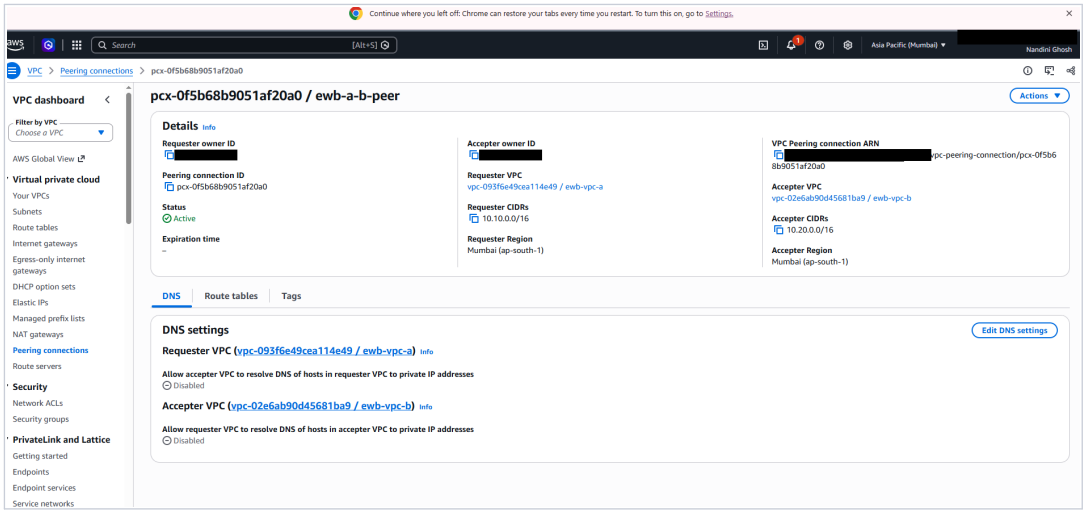


ewb-rt-b-private, Subnet associations tab.

3. Peering connection



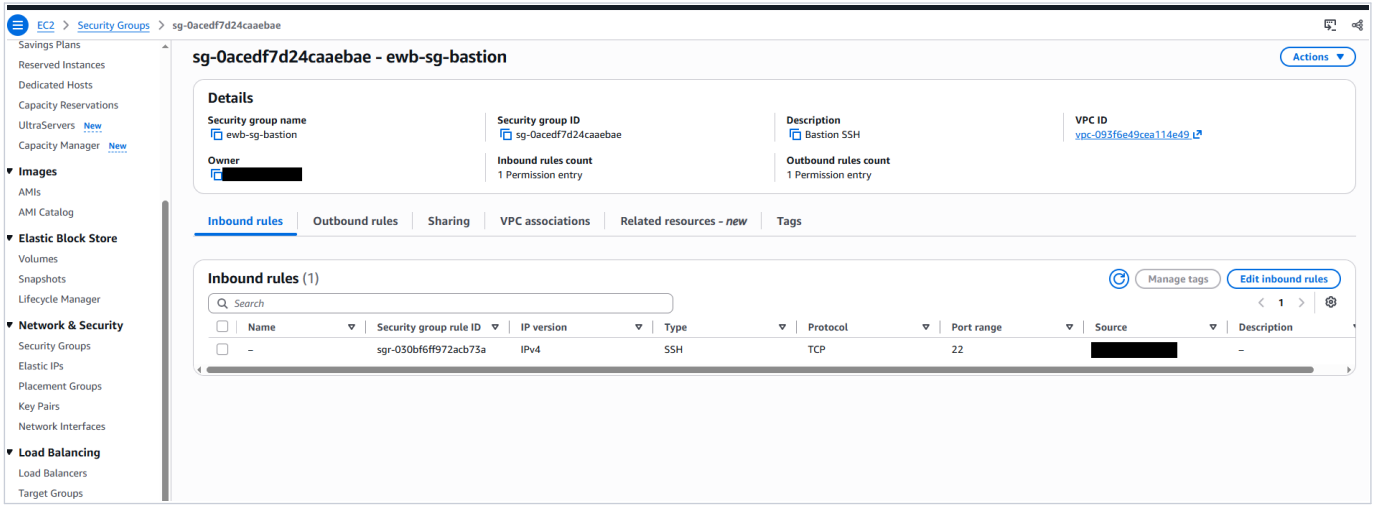
ewb-a-b-peer (pcx-0f5b68b9051af20a0): Active, requester 10.10.0.0/16, accepter 10.20.0.0/16.



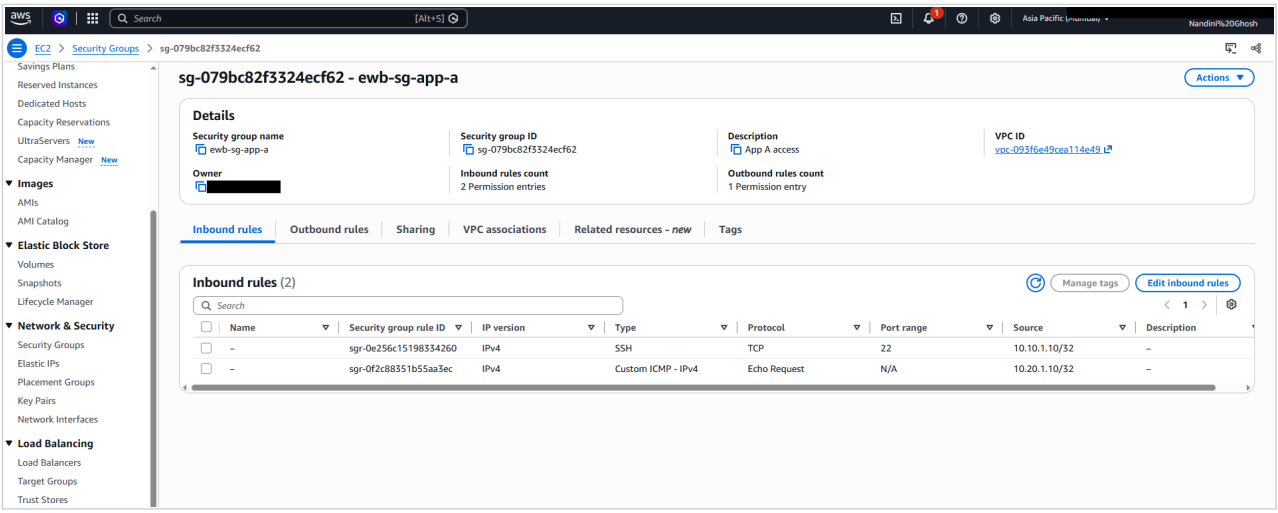
Peering connection details: status Active, requester VPC ewb-vpc-a, accepter VPC ewb-vpc-b, both in Mumbai (ap-south-1).

4. Security group inbound rules

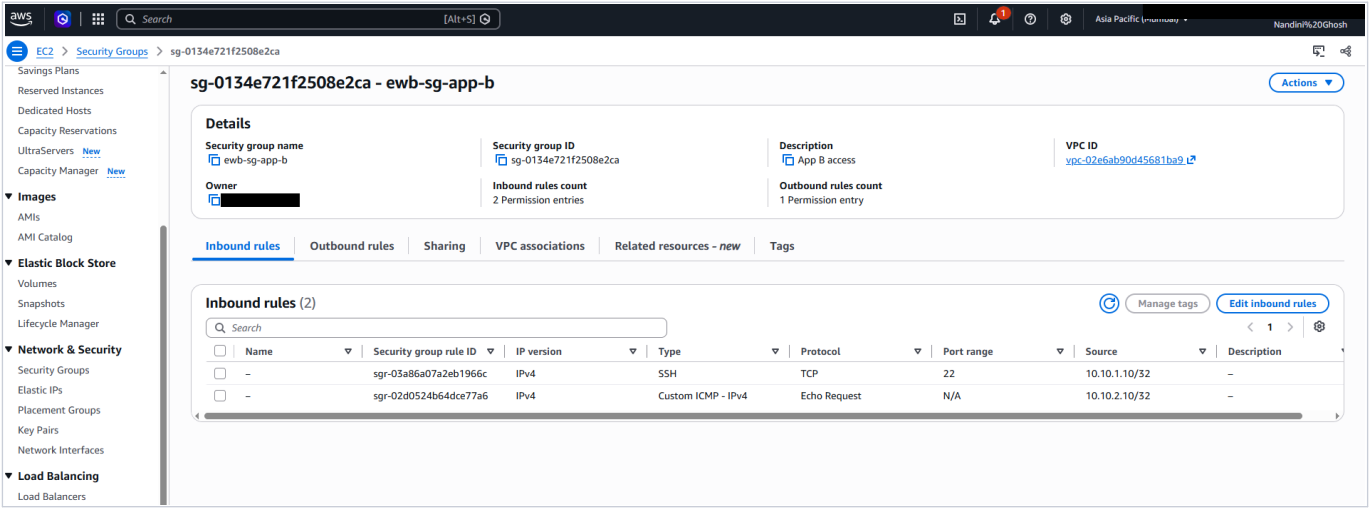
Security group	Rule	Source
ewb-sg-bastion	SSH / TCP 22	My public IPv4 /32
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-a	Custom ICMP, Echo request	10.20.1.10/32 (App-B)
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-b	Custom ICMP, Echo request	10.10.2.10/32 (App-A)



ewb-sg-bastion: inbound SSH TCP 22 from my public IP /32 only.



ewb-sg-app-a: SSH from 10.10.1.10/32 and ICMP Echo request from 10.20.1.10/32.



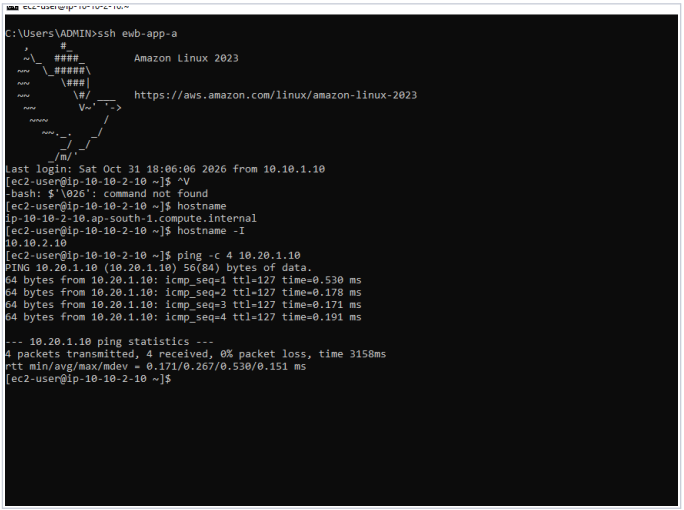
ewb-sg-app-b: SSH from 10.10.1.10/32 and ICMP Echo request from 10.10.2.10/32 (restored).

5. Test evidence

Test 01: Baseline (before peering)

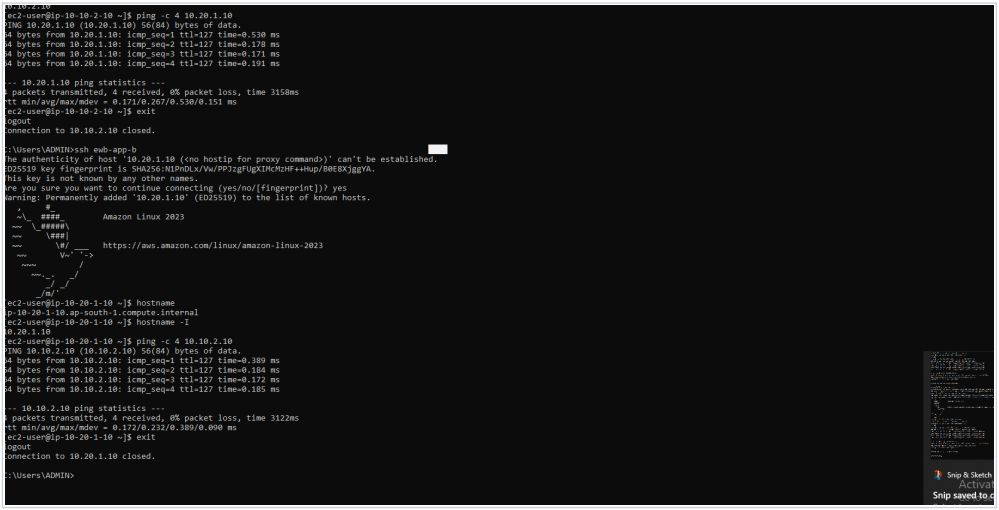
Not captured. The peering and routes were configured before I saved a baseline ping from App-A to App-B, so this screenshot is not included.

Test 02: Forward path (App-A to App-B)



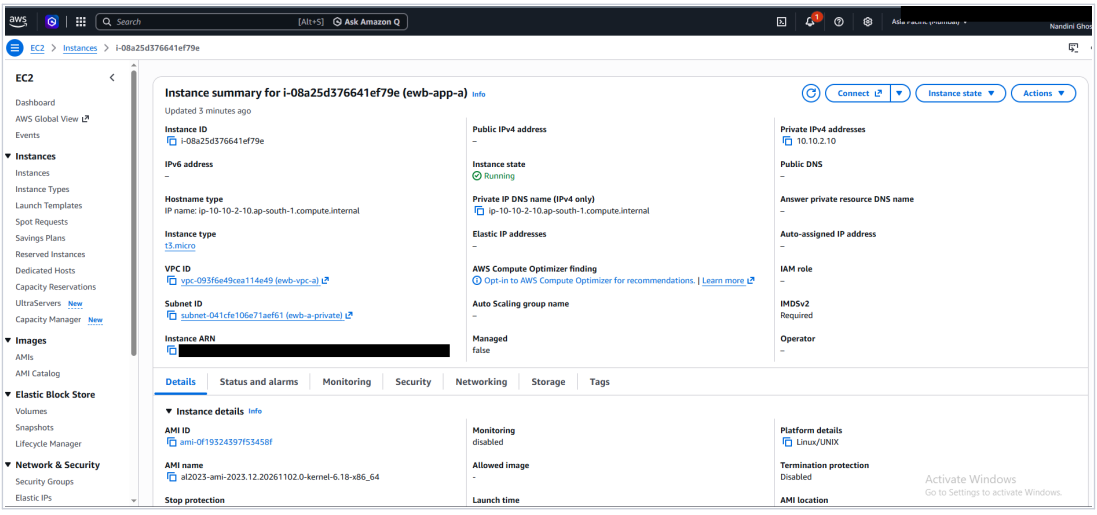
App-A (10.10.2.10), Test 02: ping -c 4 10.20.1.10 returned 4 received, 0% packet loss.

Test 03 and Test 04: Reverse path and private SSH to App-B

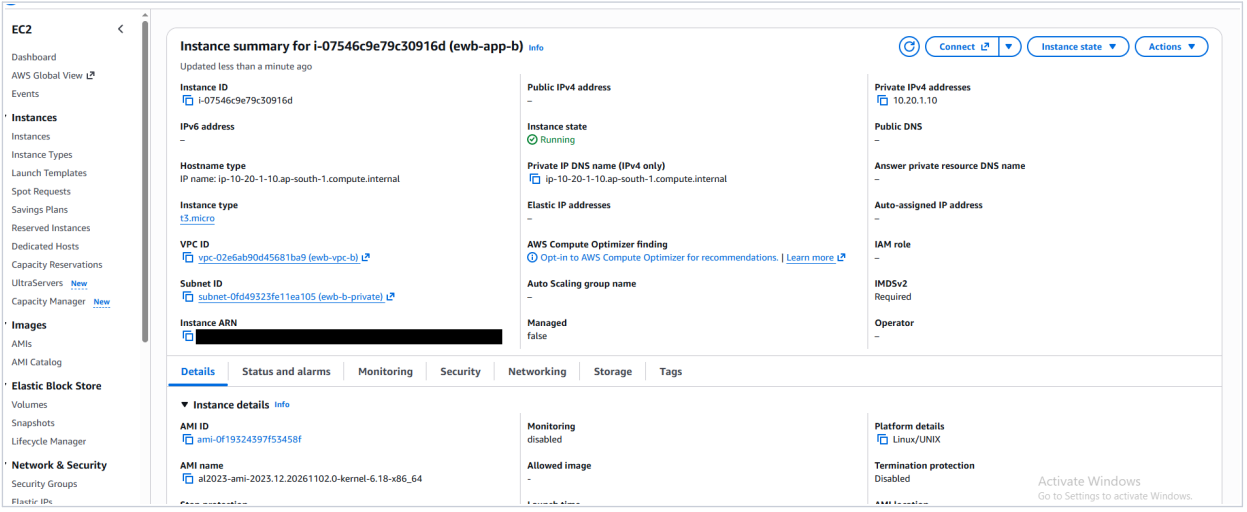


Laptop to App-B through the bastion (Test 04), then App-B (10.20.1.10), Test 03: ping -c 4 10.10.2.10 returned 4 received, 0% packet loss.

Test 05: Isolation



ewb-app-a (i-08a25d376641ef79e), Test 05: Public IPv4 address empty, private IP 10.10.2.10.



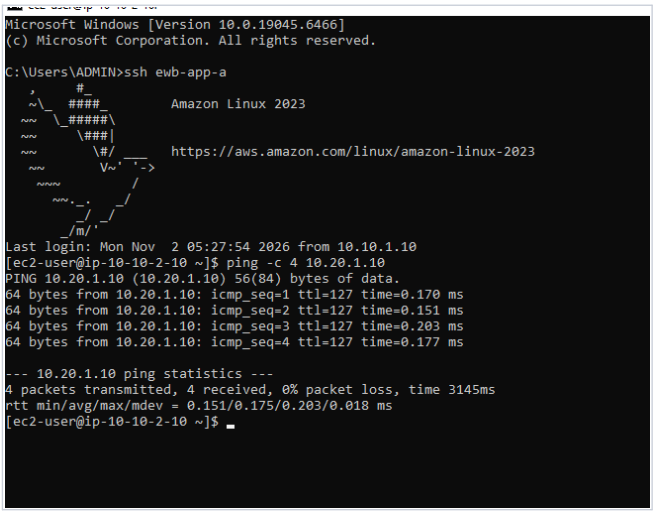
ewb-app-b (i-07546c9e79c30916d), Test 05: Public IPv4 address empty, private IP 10.20.1.10.

6. Controlled failure and recovery

I removed only the ICMP Echo request rule (source 10.10.2.10/32) from ewb-sg-app-b and kept the SSH rule.



App-A, controlled failure: ping -c 4 10.20.1.10 returned 0 received, 100% packet loss.



App-A, recovery: after the exact ICMP rule was restored in ewb-sg-app-b, ping -c 4 10.20.1.10 returned 4 received, 0% packet loss.

The pings stopped because ewb-sg-app-b no longer allowed ICMP from App-A, even though the peering and routes were still working. After I restored the exact rule, replies resumed. This shows that routes and security groups are separate controls and both must allow the traffic.

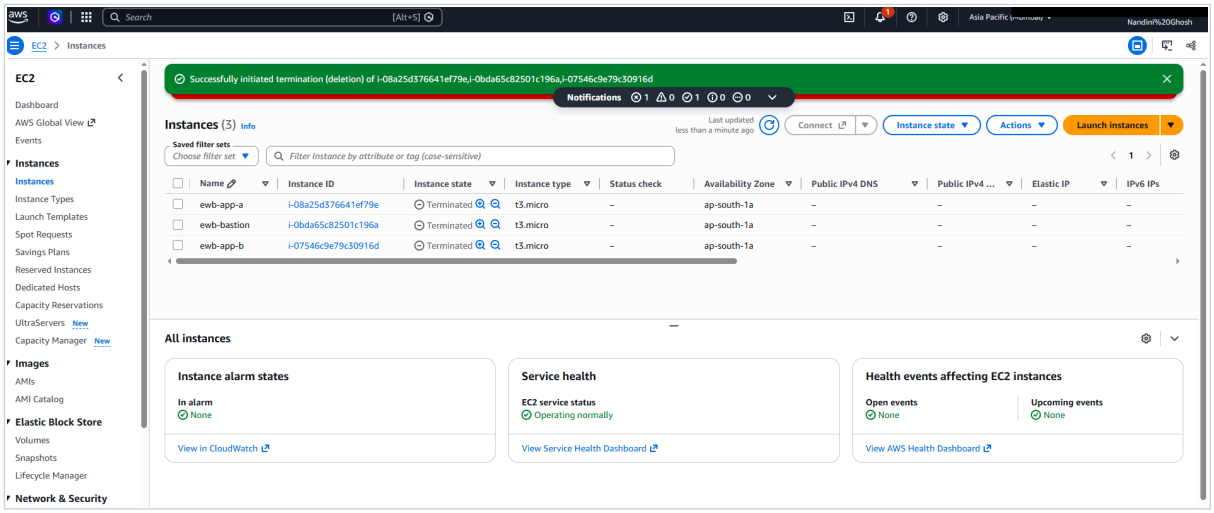
7. Forward path and return path

When App-A (10.10.2.10) pings App-B (10.20.1.10), the route 10.20.0.0/16 to the peering connection in ewb-rt-a-private sends the packet across the peering link. App-B accepts it because ewb-sg-app-b allows ICMP Echo request from 10.10.2.10/32. The reply needs a way back, so ewb-rt-b-private has the route 10.10.0.0/16 to the peering connection. Security groups are stateful, so the reply is allowed automatically. Without the return route, the request would arrive but the reply would never reach App-A.

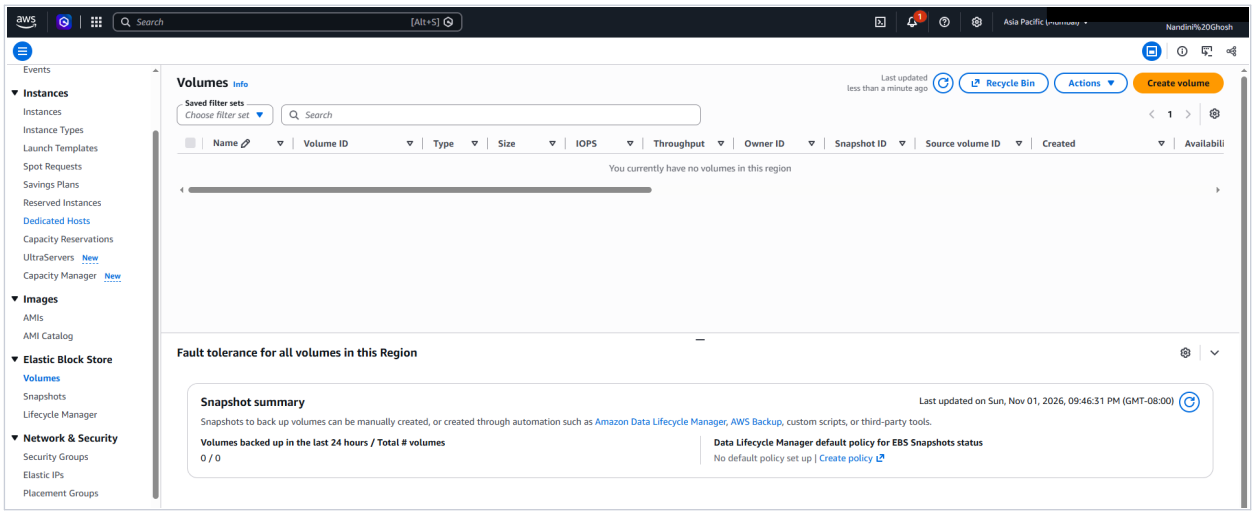
For SSH, the bastion opens a separate connection to App-B. Its subnet therefore needs the route 10.20.0.0/16 to the peering connection in ewb-rt-a-public. App-B allows SSH from the bastion private IP, 10.10.1.10/32, because that is the source address App-B sees.

8. Cleanup evidence

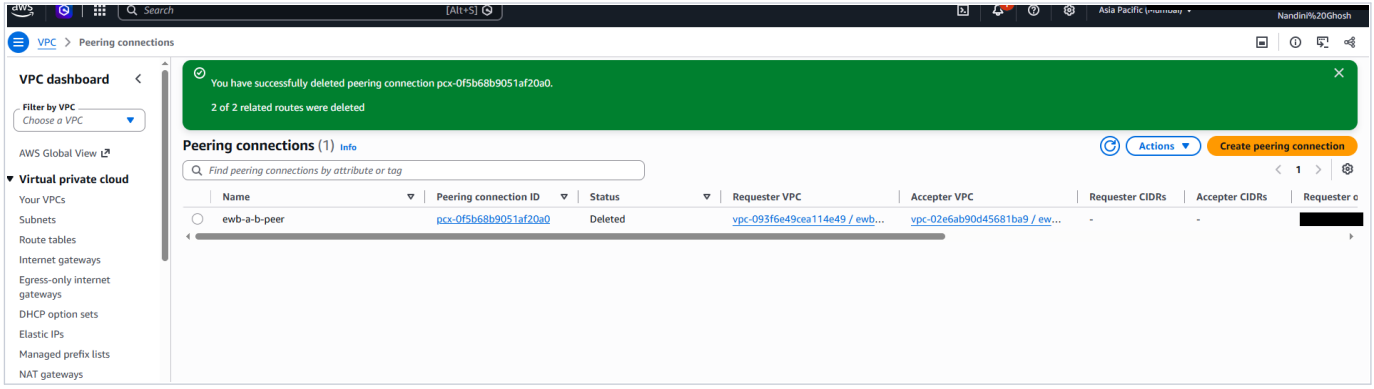
All lab resources were deleted after testing to avoid ongoing charges.



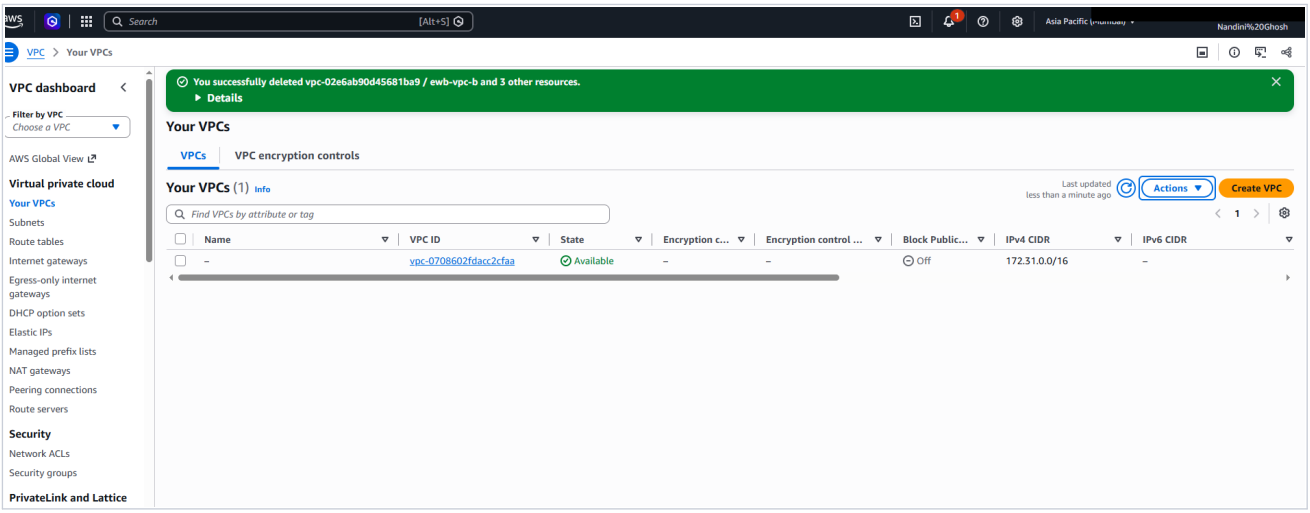
All three instances (ewb-app-a, ewb-bastion, ewb-app-b) terminated.



EC2 Volumes: no volumes remain in the region.



Peering connection pcx-0f5b68b9051af20a0 deleted, along with its 2 related routes.



Your VPCs after cleanup: only the default VPC (172.31.0.0/16) remains.

9. What I learned

Active peering alone does not make traffic flow. Both sides need routes to the other VPC's CIDR, and the security groups must allow the traffic. I learned why the return route matters, why the bastion is the only host with a public IP, and how to prove a result with a failed test, a successful test and a controlled failure with recovery.